A mixed-derivative obstruction to Weak C6 for the three-dimensional gradient

Candidate counterexample for arXiv:2412.09987, Question 6.4

19 July 2026

Status: candidate counterexample, likely valid. The displayed representation in Question 6.4 does not admit property Weak C6. This is a full negative answer to that precise question, subject to human review; it is not a solution of the section's broader representation problem.

1 Source and original question

The source is Wen Qi Zhang, *Stein-Weiss, and power weight Korn type Hardy-Sobolev Inequalities in L^1 norm*, arXiv:2412.09987v2, Section 6, Question 6.4, page 18 [1]. The question asks whether the following specific representation of ∇ on $\mathbb{R}^3$ admits property Weak C6:

$$L(D) = \begin{bmatrix} \partial_{y_2}\partial_{y_3} & \partial_{y_1}\partial_{y_3} & -2\partial_{y_1}\partial_{y_2} \end{bmatrix}, \qquad K(y) = \begin{pmatrix} y_2 y_3 \\ y_1 y_3 \\ -\frac{1}{2} y_1 y_2 \end{pmatrix}, \tag{1}$$

with Green function

$$G(x) = \nabla_x \Phi(x), \qquad \Phi(x) = \frac{1}{4\pi|x|}. \tag{2}$$

The source then proposes three cubic columns and asks whether they can be used to construct the required factorization.

2 Weak C6 in this representation

Here $V = F = \mathbb{R}$ and $E = \mathbb{R}^3$. The source definition asks for a finite-dimensional space M and homogeneous maps $T(x) : M \to \mathbb{R}$, $P(y) : \mathbb{R} \to M$, independent of the multiindex, such that

$$\sum_{|\alpha|=2} \partial_y^\alpha \big(T(x)P(y)\big) L_\alpha = \sum_{|\alpha|=2} \sum_{e_i \le \alpha} \binom{\alpha}{e_i} \partial_{x_i} G(x)\, \partial_y^{\alpha - e_i} K(y) L_\alpha \tag{3}$$

as maps $E \to V$, for every $x \neq 0$.

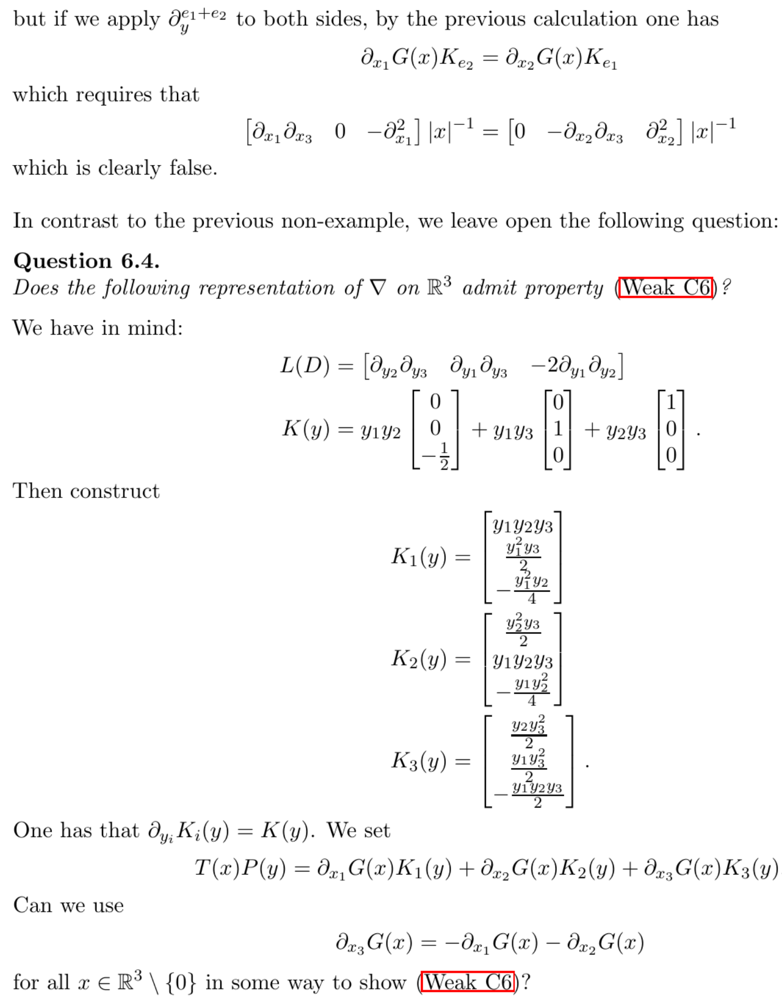
The only nonzero coefficient rows of L are

$$L_{23} = (1, 0, 0), \qquad L_{13} = (0, 1, 0), \qquad L_{12} = (0, 0, -2). \tag{4}$$

For fixed $x \neq 0$, write

$$Q_x(y) := T(x)P(y), \qquad H(x) := D^2\Phi(x), \qquad g_i(x) := \partial_{x_i} G(x).$$

Thus $g_i = (H_{i1}, H_{i2}, H_{i3})$.


but if we apply $\partial_y^{e_1+e_2}$ to both sides, by the previous calculation one has

$$\partial_{x_1} G(x) K_{e_2} = \partial_{x_2} G(x) K_{e_1}$$

which requires that

$$\begin{bmatrix}\partial_{x_1}\partial_{x_3} & 0 & -\partial_{x_1}^2\end{bmatrix} |x|^{-1} = \begin{bmatrix}0 & -\partial_{x_2}\partial_{x_3} & \partial_{x_2}^2\end{bmatrix} |x|^{-1}$$

which is clearly false.

In contrast to the previous non-example, we leave open the following question:

Question 6.4.
Does the following representation of ∇ on $\mathbb{R}^3$ admit property (Weak C6)?

We have in mind:

$$L(D) = \begin{bmatrix}\partial_{y_2}\partial_{y_3} & \partial_{y_1}\partial_{y_3} & -2\partial_{y_1}\partial_{y_2}\end{bmatrix}$$

$$K(y) = y_1 y_2 \begin{bmatrix}0\\0\\-\frac{1}{2}\end{bmatrix} + y_1 y_3 \begin{bmatrix}0\\1\\0\end{bmatrix} + y_2 y_3 \begin{bmatrix}1\\0\\0\end{bmatrix}.$$

Then construct

$$K_1(y) = \begin{bmatrix} y_1 y_2 y_3 \\ \frac{y_1^2 y_3}{2} \\ -\frac{y_1^2 y_2}{4} \end{bmatrix}$$

$$K_2(y) = \begin{bmatrix} \frac{y_2^2 y_3}{2} \\ y_1 y_2 y_3 \\ -\frac{y_1 y_2^2}{4} \end{bmatrix}$$

$$K_3(y) = \begin{bmatrix} \frac{y_2 y_3^2}{2} \\ \frac{y_1 y_3^2}{2} \\ -\frac{y_1 y_2 y_3}{2} \end{bmatrix}.$$

One has that $\partial_{y_i} K_i(y) = K(y)$. We set

$$T(x)P(y) = \partial_{x_1} G(x) K_1(y) + \partial_{x_2} G(x) K_2(y) + \partial_{x_3} G(x) K_3(y).$$

Can we use

$$\partial_{x_3} G(x) = -\partial_{x_1} G(x) - \partial_{x_2} G(x)$$

for all $x \in \mathbb{R}^3 \setminus \{0\}$ in some way to show (Weak C6)?

Figure 1: Question 6.4 and the proposed construction, from page 18 of the source paper. The crop retains the full page width and the complete question.

3 Proof intuition

Equation (3) prescribes three mixed second derivatives of the *same scalar function* Q_x. Therefore the first two prescriptions must agree after one more derivative: both become $\partial_{123} Q_x$. The right-hand sides imposed by (1), however, demand incompatible values. At $x = (1, 0, 0)$, the coefficient seen by $\partial_1 \partial_{23} Q_x$ is $-1/(8\pi)$, while the coefficient seen by $\partial_2 \partial_{13} Q_x$ is $5/(8\pi)$. No choice of the auxiliary space or factorization can alter this compatibility requirement.

4 The negative answer

Theorem 1. *The representation* (1)–(2) *of* ∇ *on* $\mathbb{R}^3$ *does not admit property* Weak C6. *Equivalently, there is no finite-dimensional* M *and no pair of maps* T, P *satisfying the source's Weak C6 identity for every* $x \in \mathbb{R}^3 \setminus \{0\}$.

Proof. Assume such M, T, P exist, and fix $x \neq 0$. Since the three rows in (4) occupy different coordinates, equality of the row maps in (3) is equivalent, component by component, to

$$\partial_{23} Q_x = R_{23} := g_2 \partial_3 K + g_3 \partial_2 K, \tag{5}$$
$$\partial_{13} Q_x = R_{13} := g_1 \partial_3 K + g_3 \partial_1 K, \tag{6}$$
$$\partial_{12} Q_x = R_{12} := g_1 \partial_2 K + g_2 \partial_1 K. \tag{7}$$

The factor -2 in L_{12} occurs on both sides and cancels in the third identity. All multiindex binomial coefficients above are one.

From (1),

$$\partial_1 K = \begin{pmatrix} 0 \\ y_3 \\ -y_2/2 \end{pmatrix}, \qquad \partial_2 K = \begin{pmatrix} y_3 \\ 0 \\ -y_1/2 \end{pmatrix}, \qquad \partial_3 K = \begin{pmatrix} y_2 \\ y_1 \\ 0 \end{pmatrix}.$$

Substitution into (5) and (6) gives

$$R_{23}(y) = \left(H_{22} - \tfrac{1}{2} H_{33}\right) y_1 + H_{21} y_2 + H_{31} y_3, \tag{8}$$
$$R_{13}(y) = H_{12} y_1 + \left(H_{11} - \tfrac{1}{2} H_{33}\right) y_2 + H_{32} y_3. \tag{9}$$

Mixed distributional derivatives commute. Hence the first two identities must satisfy

$$\partial_1 R_{23} = \partial_1 \partial_{23} Q_x = \partial_2 \partial_{13} Q_x = \partial_2 R_{13}.$$

Using (8)–(9), this necessary condition is

$$H_{22} - \tfrac{1}{2} H_{33} = H_{11} - \tfrac{1}{2} H_{33}, \quad \text{or simply} \quad H_{22} = H_{11}.$$

On the other hand, direct differentiation of Φ gives

$$H_{ij}(x) = \frac{3 x_i x_j - |x|^2 \delta_{ij}}{4\pi |x|^5}. \tag{10}$$

At $x = (1, 0, 0)$, formula (10) gives

$$H_{11} = \frac{2}{4\pi}, \qquad H_{22} = -\frac{1}{4\pi},$$

contradicting the necessary condition. Therefore no such factorization exists. □

5 Verification notes

The obstruction is independent of polynomiality. Only the second derivatives appearing in Weak C6 are required. Once they equal the smooth linear functions R_{23} and R_{13}, differentiating in the distributional sense is legitimate, so no unmentioned C^3 hypothesis on P is needed.

The proposed cubic also fails directly. The exact symbolic checker expands the source's proposed $Q_x = g_1 K_1 + g_2 K_2 + g_3 K_3$. Its residuals in (5)–(7) are respectively

$$H_{11} y_1, \qquad H_{22} y_2, \qquad -\tfrac{1}{2} H_{33} y_3.$$

This calculation diagnoses the proposed route, while Theorem 1 rules out all other routes for the same G, L, K.

The suggested differential identity is false. The source asks whether one can use $\partial_{x_3} G = -\partial_{x_1} G - \partial_{x_2} G$. At $x = (1, 0, 0)$, the three rows of $D^2\Phi$ show immediately that this identity does not hold. The proof of Theorem 1 does not depend on this observation.

Computational audit. Run

```
conda run --no-capture-output -n sandbox python \
  code/check_weak_c6.py
```

from this packet directory. The script uses exact symbolic arithmetic to expand all three prescribed right-hand sides, check the mixed-derivative obstruction, show that a general homogeneous cubic is inconsistent at the chosen point, and audit the paper's tentative cubic. The computation is not a substitute for the proof.

6 Scope, novelty check, and review recommendation

The theorem fully answers the exact displayed Question 6.4 in the negative. It does not settle the broader question in Section 6 of whether ellipticity and cancellation imply the existence of *some* Weak C6 representation, nor does it exclude other choices of L and K for ∇.

Before promotion, the run's lightweight registry, solution, and attempt indexes were searched using arXiv:2412.09987, the exact phrase "Weak C6", "WTP", and core gradient-representation terms. A bounded arXiv/web search used the exact question phrase, the source title and id, and close Weak C6 terms. It found the source paper, whose current v2 is dated 15 May 2025 and still poses Question 6.4, and no separate answer. This records the search bounds and is not an exhaustive novelty claim.

Human review is recommended. The two most important checks are (i) the componentwise extraction of (5)–(7) from Weak C6, including the composition order and the -2 coefficient, and (ii) the mixed-derivative compatibility step. Both are explicit and local.

References

[1] W. Q. Zhang, *Stein-Weiss, and power weight Korn type Hardy-Sobolev Inequalities in L^1 norm*, arXiv:2412.09987v2, 2025. `https://arxiv.org/abs/2412.09987`.